

AcceloPrint: Fabricating Customizable Accelerometers with Multi-Material 3D Printing

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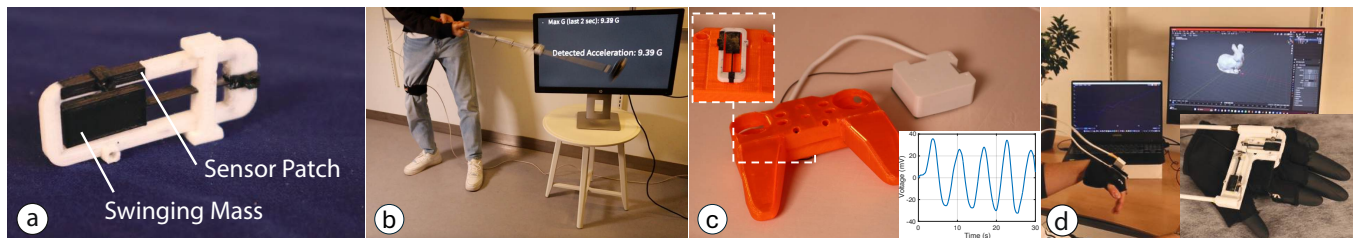


Figure 1: *AcceloPrint* sensors are fully 3D printed acceleration sensors. They consist of a swinging mass and a sensor patch (a) Acceleration tracking of a golf club with *AcceloPrint* sensors (b). Tight fit placement of the *AcceloPrint* sensors on the gamepad and cable connections from the *AcceloPrint* board via screws (c) and as tangible tools for interaction such as a 3D mouse (d).

Abstract

We introduce *AcceloPrint*, 3D-printed acceleration sensors that can be fabricated in one pass alongside a 3D object and report on its angular orientation or acceleration. *AcceloPrint* utilizes capacitive sensing to track the deflection of a 3D printed cantilever beam to a sensor patch. Our *AcceloPrint* tool integrated into a 3D editor generates a sensor with a user-defined sensing range generated by our computational model. We also propose a novel sensor design with an adjustable sensing range post-fabrication. Our technical evaluation shows our sensor can detect acceleration up to 50 m/s^2 , with a root mean squared error of 0.35 m/s^2 (%3.57) in the range up to 10 m/s^2 . We demonstrate *AcceloPrint* with three application examples on sports performance tracking and tangible tools.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI).

Keywords

fabrication, sensors, tangible user interfaces, 3D printing, multi-material printing

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1 Introduction

3D printing has evolved from a tool for creating simple prototypes to a versatile technology capable of fabricating increasingly complex objects at low costs. While many 3D-printed objects have traditionally been non-interactive, researchers demonstrated methods to embed interactive elements, such as capacitive touch sensors [18], bend sensors [20], and rotary encoders [1] directly into the fabrication process. This growing library of 3D-printable sensors promises the opportunity for makers and researchers to create custom tangible user interface with integrated sensors and output components.

For accelerations sensors, researchers explored 3D printing deformable beams [12] and springs [3, 10, 11] that bend or compress under acceleration. The deformation can be detected via capacitance. However, the conductive electrodes for sensing capacitance have to be manually attached or screen-printed onto the surface of the 3D-printed geometry. Printed Optics [22] embeds optical fibers into a 3D print that can deform under acceleration and its displacement can be sensed with an IR receiver. Closest to our research is a fully 3D printed accelerometer presented in MetaSense [6]. The sensor is embedded into a mechanical metamaterial and uses capacitive sensing of a printed electrode pair to sense the deformation of a beam similar to *AcceloPrint*. However, both Printed Optics and MetaSense do not present a characterization of the sensor and it is unclear how well they perform.

We introduce *AcceloPrint*, a fully 3D printed acceleration sensors using multi-material 3D printing. The printed sensors do not require any assembly or removal of support material and, thus, are ready to be used directly after printing. Their fabrication takes less than one hour at a cost below \$1.5. While *AcceloPrint* is not intended to serve as a direct replacement for commercial accelerometers, it provides a low-cost, rapidly fabricated alternative for prototyping

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and exploratory applications, particularly in research and early-stage development. We propose a sensor layout that utilizes the capacitive tracking of a swinging mass to sense the acceleration of an object. The *AcceloPrint* sensor can detect acceleration and tilt on one axis or, by combining three sensors, all three spatial dimensions. We present a computational tool that lets users define a custom sensitivity to track either fine acceleration difference at high precision (e.g. when only tilting an object) or strong accelerations (e.g. for tracking the acceleration of a golf club). Our *AcceloPrint* design tool automatically generates the sensor geometry according to the user's acceleration sensitivity preferences in a 3D editor. Our technical evaluation shows that *AcceloPrint* sensors have a sensing accuracy between 3.6% (sensing range: $10m/s^2$) and 10.6% (sensing range: $260m/s^2$), and we demonstrate the reliability of our sensor between multiple prints. We show the usefulness of *AcceloPrint* sensors with three application examples.

Taken together, we contribute:

- a *sensor design* of a 3D printable acceleration sensor with fixed or variable sensing range based on capacitive sensing;
- a *3D editor extension* that automatically generates *AcceloPrint* sensors with a user-defined sensitivity using our computational model and a Java library that converts the raw sensor data into acceleration values;
- a technical evaluation of the sensing accuracy for multiple prints up to $50m/s^2$.

2 Related Work

This work builds on the growing body of research on fabricating custom sensor and tangible user interfaces. We position our contributions within research advances on fabricating accelerometers in engineering and material science, 3D printed sensors in HCI, and fabrication approaches for tangible user interfaces.

2.1 3D Printing Acceleration Sensors

Similar to *AcceloPrint*, researchers utilize capacitive sensing of a deformable beam to detect acceleration. Baek et al. [3] integrated a copper coil into a soft silicone rod that tracks its bend via capacitive sensing. Liu et al. [10, 11] show a fully 3D printed accelerometer consisting of multiple parts that have to be manually layered on top of each other. Zega et al. [23, 24] show a fully 3D printed 3-axis accelerometer geometry using springs that can be printed in one pass requiring only the removal of support material. Zega et al. [25] later demonstrated a miniaturized spring-based accelerometer using laser powder bed fusion (L-PBF) printing with manual attachment of a copper electrode. Barile et al. [4] 3D printed a similar structure but used conductive PLA for the swinging mass which only requires to manually attach copper plates. MetaSense [6] utilizes modular meta-material structures to embed capacitive sensors in deformable objects to sense motion and acceleration. Printed Optics [22] embeds optical fibers into a 3D print that can deform under acceleration which can be sensed with an IR emitter/receiver pair. Both MetaSense [6] and Printed Optics [22] show a promising method for printing acceleration sensors, however, the sensors are not characterized and it is not clear how well they perform. Another stream of research utilizes piezo-resistive sensing on deformable beams. Liu et al. [12] 3D printed a deformable beam on which they

screen printed a U-shaped sensor with carbon-conductive paint. The conductive layer bends with the 3D printed structure under acceleration which results in a measurable resistance change. Closest to our research is Arh et al. [2] which follows the same principle but fully 3D prints the sensor using conductive carbon filament. In contrast to *AcceloPrint*, the sensing range is fixed after printing and it is not clear how well the sensor performs between multiple prints. *AcceloPrint* builds on this previous work and contributes a fully 3D printed design that does not require any manual post-processing after fabrication.

2.2 3D-Printed Interfaces in HCI

Sensing user input and providing output to the user is a core challenge for 3D printing tangible user interfaces. Similar to our sensing method, researcher used capacitive sensing to detect interaction. PrintPut [5] and Capricate [18] demonstrate a digital design tool that facilitates the integration of touch sensitive areas and internal wiring in 3D printable objects. Flexibles [20] propose to utilize the touch sensors of a tablet and 3D printed conductive structures in the interior of soft objects to detect their deformation. CAPatch [9] extends this approach with tangible sliders and knobs. itsy-bits [19] proposes a unique footprint on tangibles that allow the capacitive touch surface of a tablet to identify individual tangibles through this footprint. Alalawi et al. [1] show in MechSense that capacitive sensors can be used to create fully 3D printed rotary encoders in rotational mechanism such as gears and joints. Ozbek et al. [14] show a fully 3D printed soft pressure sensing spring that can be integrated into a robot to detect obstacles and collisions while moving.

AcceloPrint adds to this body a work a 3D printed acceleration sensor that can be printed in one pass with customizable sensing range. Our work aims at adding to the already existing library of 3D printed sensors a novel sensor type that can be used to create more versatile tangible user interfaces via digital fabrication tools.

2.3 Adding Electronic Components to Objects

HCI researchers also explored how to attach or embed off-the-shelf electronics inside and on objects. Early work focused on augmenting objects with conductive copper tape for capacitive touch sensing (Midas [17]) or full circuitry with additional electronic components attached to the surface of these objects (SurfCuit [21]). ModElec [7] extends this work with a digital tool that also allows to integrate electric components onto the 3D printing traces. MorphSensor [27] present a digital tool to deconstruct ICs and distribute individual electric components on the surface of objects. Maker's Marks [15] simplifies the placement of electronic components by letting users place stickers as placeholders for the components on an objects that is subsequently 3D scanned and 3D printed with holders for these electrical components. 'A series of Tubes' [16] leaves empty space in 3D prints to fill these 'tubes' with several interactive materials, e.g., thin electroluminescent displays for visual feedback. In a similar way, Curveboards [26] embed breadboard structures in 3D printed objects that feature internal tubes filled with conductive silicone and allows to place and prototype with electronic components on the surface of objects. LaserFactory [13] further automates the placement of electronics via a modified laser cutter that can paint silver traces, place electrical components, and sinter the silver

traces to create fully functional devices without user intervention. *AcceloPrint* complements this line of work by fully 3D printing an acceleration sensor that can be customized for specific acceleration intensities by the maker for creating tangible user interfaces.

3 AcceloPrint

AcceloPrint is a fully 3D-printed acceleration sensor that supports customizable sensing ranges. Below, we outline the working principle, design geometry, and customization options.

3.1 Working Principle

AcceloPrint detects acceleration by measuring capacitance changes in a capacitive plate located parallel to a swinging pendulum. The sensor consists of three main components: a frame, a sensing plate, and a swinging mass (Figure 2). The frame and sensing plate remain stationary, while the swinging mass deforms under acceleration, altering its distance from the sensing plate. The system is fabricated using conductive PLA (cPLA) for the sensing plate and swinging mass, while the frame is made from non-conductive PLA. Changes in capacitance occur due to variations in the distance between the sensing plate and the swinging mass as it deforms in response to acceleration (Figure 2c).

3.2 Customization for Sensing Ranges

The stiffness of the swinging mass is primarily determined by the beam's length and width. A longer beam increases flexibility, enhancing sensitivity to finer accelerations, while a shorter beam provides greater stiffness, allowing for the measurement of higher acceleration ranges with reduced deflection. By leveraging these properties, *AcceloPrint* enables two distinct customization methods:

- (1) **Adjustable Sensor:** A mechanical slider allows users to modify the beam length after fabrication, providing real-time adaptability for different sensing ranges (Figure 2a). By extending the beam, the sensor becomes more sensitive to small accelerations, while shortening it increases stiffness for detecting higher accelerations. This configuration is ideal for applications requiring on-the-fly tuning without reprinting the sensor.
- (2) **Static Sensor:** In contrast, a fixed beam length is predefined based on the desired sensing range using the *AcceloPrint* design tool (Figure 2b). The design tool maintains all beam parameters constant such as width and thickness while only varying the beam length to achieve different sensitivity levels. This approach ensures precision and repeatability in acceleration measurement, as the beam dimensions are optimized for specific application requirements. The static design is best suited for scenarios where stability and consistency are prioritized over post-fabrication adjustability.

3.3 Sensing Capacitance and Signal Processing

The system uses a 64V, 7 kHz sinusoidal AC signal applied to the swinging mass by an AC inverter. A microcontroller (ESP32) reads the resulting voltage changes at the sensing plate. We filter the raw sensor data with a maximum filter to only sense the voltage

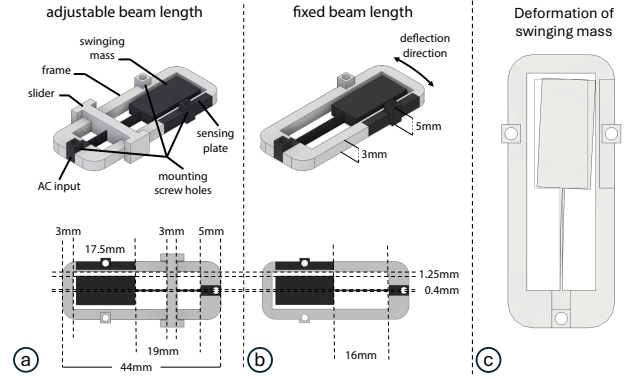


Figure 2: Sensor schematic of our adjustable slider design (a) and the fixed beam length design (b). The swinging mass is connected to an AC signal. The sensing plate measures the distance to the swinging mass using capacitive sensing. The slider can modify the beam length to adjust the sensitivity of the swinging mass. Under acceleration, the swinging mass bends towards or away from the sensing plate (c).

peaks and a Kalman filter [8] for Arduino¹ (Figure 3e). We noticed that wires between the sensor and the microcontroller can be influenced by environmental noise. To improve the signal quality, we use an EM-shielded cable to connect the sensing patch to the microcontroller (coaxial cable² with shielding connected to the ground).

3.4 Sensor Calibration

We found in our wearable application example that the presence of noise like the human body can interfere with our sensor. Thus, we recommend to calibrate the sensors when using them as a wearable or when close to a source of interference. This is also necessary when using the slider version of our sensor. Since our system does not know the current beam length, the user has to perform a calibration so that our system can fit a custom estimation function to the current sensor configuration. While connected to our sensing board, the user rotates the sensor to 0° (horizontal) and 90° (vertical) such that there is a 0G and a 1G acceleration incident through earth's gravity. Our tool reads these voltage values and fits the quadratic function ax^2 to this value computing the value a . We use the parameter a to fit the same quadratic polynomial to all sensing values and estimate the acceleration through this function.

3.5 Sensor Integration

To facilitate a stable connection and placement on interactive devices and objects, we add three screw holes to the *AcceloPrint* sensor design. These screw holes are connected to the sensing plate and the swinging mass, respectively. The maker can use a screw to attach a wire or directly solder the wire onto the screw. During our experiments, we also found that heating jumper wires with a

¹<https://github.com/denyssene/SimpleKalmanFilter>

²Sinox - Satellite Cable 70db

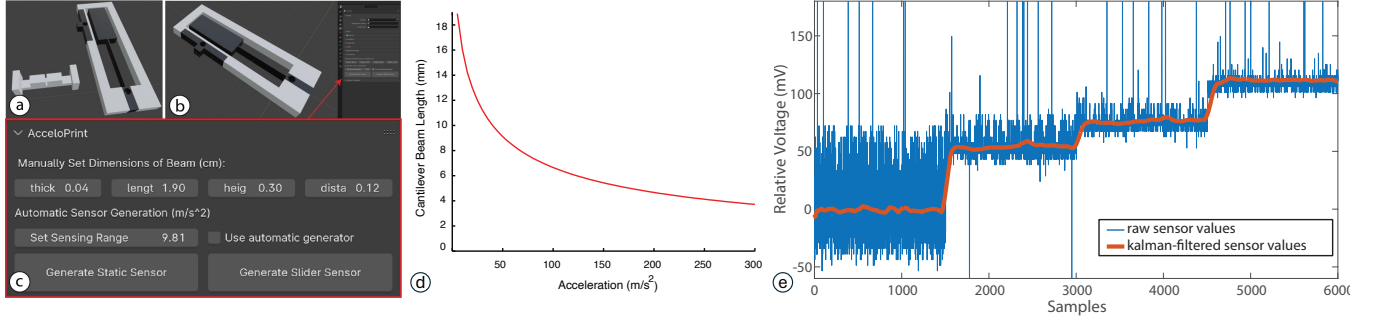


Figure 3: *AcceloPrint* plugin for Blender. Users have the option to generate a slider sensor (a) or a custom static sensor with variable sensing range using different beam lengths. They have in our tool (c) the option to manually set the beam parameters or let our system automatically set these parameters for a user-defined sensing range. (d) Graphical representation of the relationship between *AcceloPrint*'s beam length and acceleration as derived from reverse-solving the Euler-Bernoulli Equation for a fixed deflection input of 1.25mm. (e) Raw sensing values (blue) and filtered values (red). Our sensing board reads values with noise that are centered around the actual capacitance. We use a Kalman filter to estimate the incident capacitance of the system.

lighter and pressing the hot tip into the cPLA to be a quick and stable connection.

4 Fabrication

AcceloPrint sensors are designed using our Blender plugin (Figure 3) to allow customizable sensing ranges, 3D printed, and integrated with a custom sensing board.

4.1 Design and Beam Length Estimation

The Blender plugin generates print-ready geometries for *AcceloPrint* sensors. Users can create slider sensors for post-fabrication sensitivity adjustment (Figure 3a) or custom static sensors with specific sensing ranges (Figure 3b). The beam length is determined using the Euler-Bernoulli equation for cantilever beams with fixed deflection. We calculate the beam length required for a given acceleration by integrating and solving for initial conditions. Using a fixed deflection of 1.25mm, the beam length (L_{Can}) is related to acceleration (A_{max}) by:

$$L_{Can} = 40.65 \cdot A_{max}^{-0.1769} - 11.24 \quad (1)$$

This computation is embedded in the Blender plugin to generate custom sensor for a user-defined acceleration range input (Figure 3d).

4.2 3D Printing

Sensors are fabricated using a FlashForge Creator Pro 2 3D printer with conductive PLA (Protopasta cPLA) and non-conductive PLA (Elegoo White PLA). Print settings include a layer height of 0.18 mm, nozzle temperature of 210°C, and build plate temperature of 80°C. Sensors are printed onto painter's tape applied to the print bed for easy removal (Figure 4). The process involves printing, removing the sensor along with the tape, and peeling off the tape to reveal the completed sensor. The cost of the largest sensor is approximately \$1.50.

4.3 Sensing Board

The sensing board integrates an ESP32 microcontroller, an ADC (MPC3008), and an inverter to generate an AC signal from DC input. The ESP32 processes sensor data and transmits it wirelessly via ESP-NOW (Figure 4). The board design includes all necessary components for signal generation, data acquisition, and wireless communication, ensuring seamless integration with the 3D-printed sensors.

5 Technical Evaluation: Accuracy and Repeatability

We evaluated the accuracy and repeatability of *AcceloPrint* sensors using a centrifugal test setup (Figure 4f and g) to compare their performance against a commercial IMU (MPU6050). The evaluation included 25 fixed-beam sensors and 5 adjustable slider sensors tested across a range of acceleration values (0 to 50 m/s^2) and beam lengths (4mm to 16mm).

Apparatus and Procedure. Sensors were mounted at a fixed distance from the center of a rotating platform opposite the IMU for ground truth measurements. Data was collected at 100 Hz for 20 seconds per speed, discarding unstable readings during speed transitions. Calibration was performed at 10 m/s^2 to derive the estimation function ax^2 for acceleration detection.

Results: Fixed-Beam Sensors. Figure 5 shows that fixed-beam sensors detected acceleration with errors ranging from 3.6% (16mm) to 10.6% (4mm). Shorter beams showed higher errors due to smaller voltage amplitudes and increased noise sensitivity. For shorter beams (4mm and 7mm), the sensing range exceeded the test limit (up to 260 m/s^2 for 4mm). Longer beams plateaued as the swinging mass touched the frame, indicating the maximum sensing range.

Results: Slider Sensors. The slider variant showed similar trends, with errors ranging from 3.8% (16mm) to 9.9% (4mm). However, slider sensors were more flexible due to tolerance between the slider and the beam, resulting in slightly reduced sensing ranges for the same beam lengths compared to fixed-beam sensors.

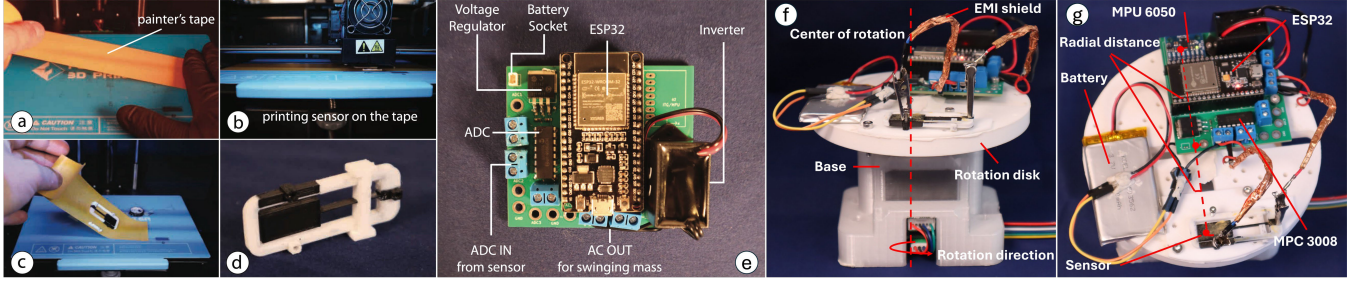


Figure 4: The process of FDM 3D printing of the *AcceloPrint*. (a) We apply adhesive tape onto the 3D printer bed to serve as a removable surface for sensor printing. We printed the *AcceloPrint* sensor directly onto the tape (b) which can be peeled off from the printer bed (c) without damaging the sensor (d). (e) The *AcceloPrint* Board. We use an ESP32 microcontroller for signal processing and wireless communication (ESP-NOW). The AC signal gets generated with an inverter and we read the sensor signals with an ADC. (f) Centrifugal Test Setup. We mounted our sensor on a rotating platform to test the sensor at 0, 2.5, 5, 7.5, 10, 15, 20, 30, 40, 50 m/s^2 . The base (f) houses a stepper motor that rotates the platform (g). We placed an IMU (MPU6050) opposite the *AcceloPrint* sensor to generate a ground truth acceleration for our evaluation.

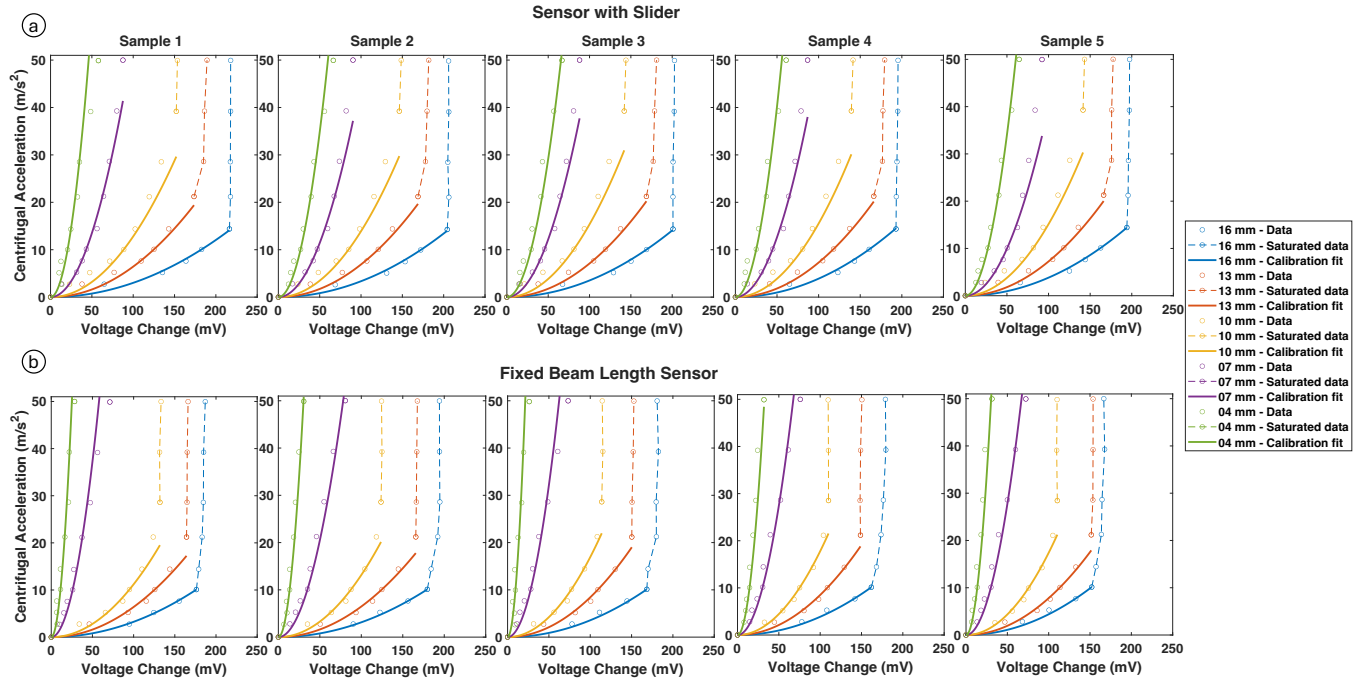


Figure 5: Results of the centrifugal acceleration experiment with the slider variant of the sensor (a) and the fixed beam length sensor (b). Our estimation function (continuous lines) matches the ground truth data with a maximum error of 10.6% for the shortest beam and a minimal error of 3.6% for the longest beam (16mm).

Repeatability Across Prints. Voltage readings were consistent across multiple prints, with standard deviations between identical samples of slider sensors ranging from 1.4% (13mm) to 3.5% (16mm) and fixed-beam sensors ranging from 1.2% (4mm) to 4.6% (16mm). These small deviations suggest that calibration can effectively mitigate differences between prints, and more precise fabrication methods could reduce the need for calibration.

6 Application Scenarios

We demonstrate *AcceloPrint* with three application examples on tangible interaction tools.

Sport Performance Tracking Tool. We demonstrate an interactive gold club utilizing an adjustable *AcceloPrint* sensor (Figure 6a). Users set the slider to a small sensing range up to 4G to practice slow swings at high precision (Figure 6e). Alternatively, they can set the slider to a high acceleration range at lower resolution

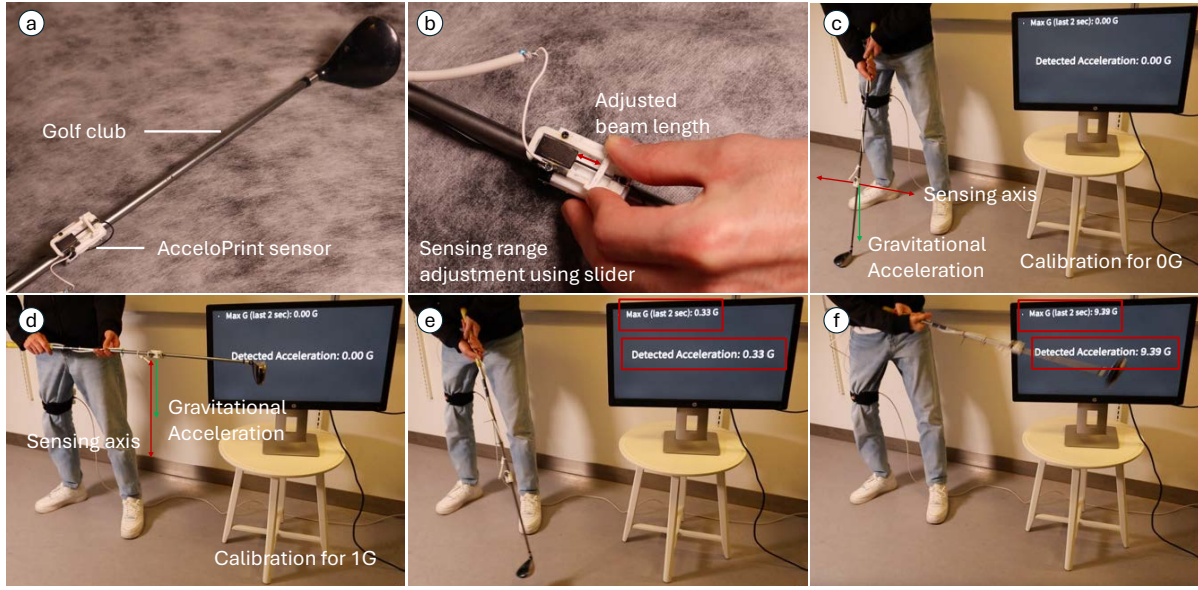


Figure 6: Sport performance tracking. We demonstrate an interactive gold club that can track the strength of a swing using an adjustable *AcceloPrint* sensor for practicing fast and slow swings (a). Setting up the sensor for various sensing ranges by adjusting beam length via slider (b). The user calibrates the sensor at 0G and 1G (c,d). After calibration system can detect both relatively low accelerations such as 0.33G (e) and relatively high accelerations such as 9.39G (f).

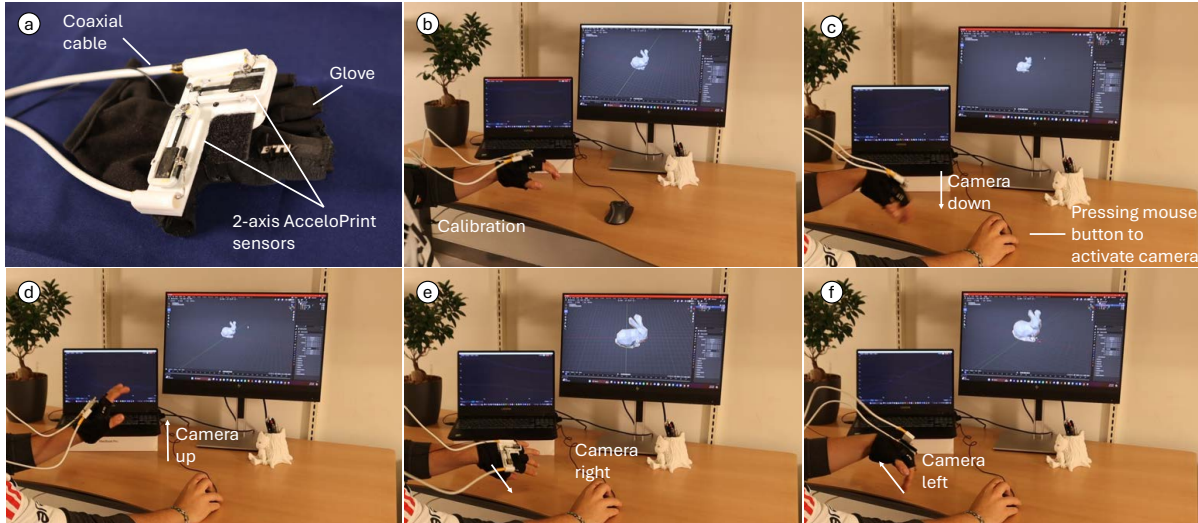


Figure 7: Tangible interaction tools. We demonstrate a 3D mouse using two *AcceloPrint* sensor mounted on a glove which can sense tilt on two axes (a). The user calibrates the sensor while wearing it (b) and can control the camera in a 3D editor by tilting their hand down (c), up (d), right (e), or left (f) to let the camera rotate and assist in more natural navigating in 3D space.

to practice strong swings (Figure 6f). After each adjustment, the sensor has to be recalibrated (Figure 6b-d).

Tilt-Aware Controller. We integrated an *AcceloPrint* sensor on a 3D printed game controller. The sensor is attached to the controller using the integrated mounting holes and can sense tilt to the sides (Figure 1c). Printing and attaching the sensor took only 25 minutes.

Interactive 3D mouse. We created a 3D-printed interactive mouse for navigating 3D models in Blender by placing two *AcceloPrint* sensors on a glove (Figure 7a). We calibrate the sensors for up, down, right, and left directions (Figure 7b). The sensor's tilt detection enables intuitive camera control, enhancing interaction with digital environments (Figure 7c-f).

7 Future Work and Conclusion

AcceloPrint demonstrates the fabrication of acceleration sensors that can be printed in one pass and at low cost enabling the detection of angular orientation and acceleration. By eliminating the need for support materials, the process requires minimal user intervention and our technical evaluation shows that the sensors accurately detect acceleration up to 50 m/s^2 , with a minimal error of 0.35 m/s^2 (3.57%) in ranges up to 10 m/s^2 . In the future, we want to explore the feasibility of directly integrating *AcceloPrint* sensor in 3D printed objects in any orientation and miniaturize our sensors for a more seamless integration in various tangible user interfaces.

Integration Pipeline. Currently, users must manually integrate our sensors into objects after fabrication. We aim to establish a fully integrated pipeline, allowing sensors to be freely placed within digital objects with automatic internal wiring generation. Key challenges include:

- (1) printing sensors in arbitrary orientations, which may require removable support materials such as water-soluble supports; or optimizing sensor geometries for printing in x-, y-, and z-orientation.
- (2) shielding internal wiring printed within objects to prevent interference. A solution could involve printing conductive shells around wires that can be grounded.

Miniaturization. Integrating electronics into compact 3D objects is challenging due to the size of components like sensors, batteries, and circuitry. While *AcceloPrint* is one of the smallest 3D-printed accelerometers compared to related work (2.1), there is potential for further miniaturization using high-conductivity filaments such as Electrifi³.

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